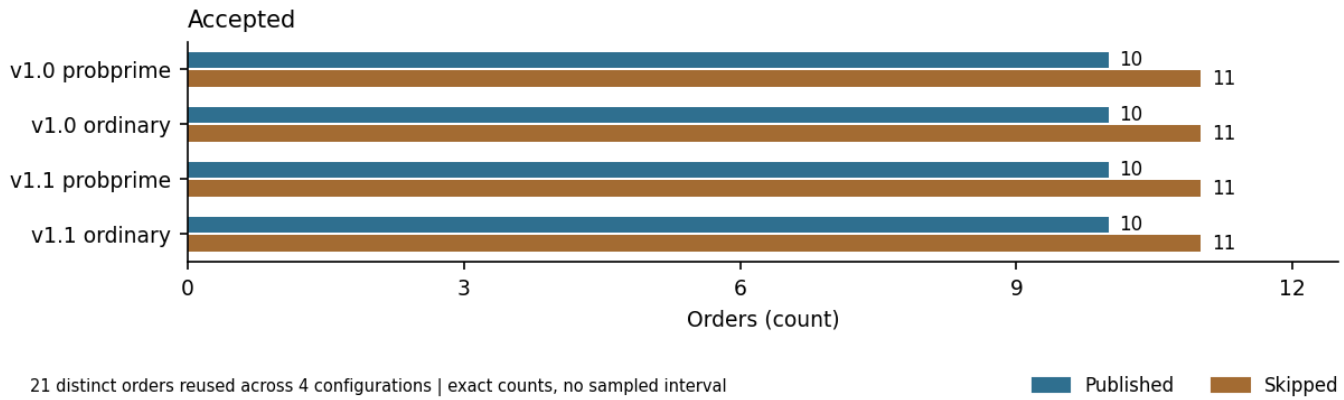
Recovered FreeLIP routines accept every published and skipped order

27 October 2026. All 21 original group orders pass both tested primality routines in both recovered FreeLIP versions. The fixed replay completed 84 calls: 10 published and 11 skipped orders per configuration, with zero software rejections and zero unknowns after source-faithful output validation. These routines do not explain the skips when given the correct group orders. The historical caller, its computed inputs and its point-counting code remain unrecovered. The original criterion goal is still unresolved.

Every fixed configuration accepts all 21 orders



The graph plots the native summary. Four configurations reuse the same 21 historical choices; they are not 84 independent historical samples. Exact counts have no sampling interval. The complete table and native unified export retain every call. Stream/index labels and counts are decimal; exact curve parameters and ICV1 identities remain in the frozen parent inputs and original assessment.

Why this library was tested

The primary curve-origin report links six May 1997 published base points to the FreeLIP RNG and seven discarded single-limb outputs. This gives FreeLIP a provenance connection to the observed generation process. It does not identify the primality routine or original caller. The independent enumeration still describes the remaining skips as unexplained. The previously tested ECPP artifact likewise had no recovered original-caller link.

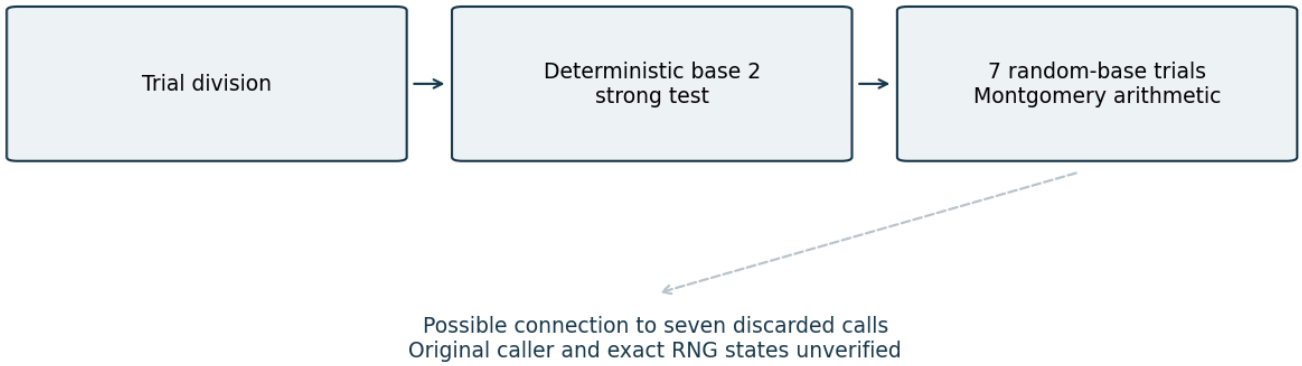
FreeLIP 1.0 was recovered from the FUNET archive and GWDG mirror. Both 201,037-byte archives have SHA256 6ef1e0bb398eec0c6cbd22f51b06cbbbbd4d995673ae8da942de6f9d3a7ecec5. The original README dates 1 May 1995. Version 1.1 comes from pinned mirror commit 00898a4814d2cc1244bf348682be6ed10dda35f0, with original README dated 21 July 1997. Source bodies and original notices are preserved. Mirror agreement binds reproducible bytes, not an authenticated 1997 copy or evidence that the curve generator used this version.

The byte-bound source audit finds seven relevant bodies identical across the recovered versions: zprobprime, zmcomposite, zsmexp, zrandom, zrstarts, zrandomb and zmstartint. The zcomposite difference is memory cleanup. The documented extended-Euclid fix in 1.1 is not by itself evidence of a primality failure, a skipped curve or a vulnerability.

A concrete seven-call mechanism, with a remaining gap

Source zprobprime(n,t) performs trial division, a deterministic base-2 strong probable-prime test, then t random-base Montgomery tests. Fixing t=7 therefore uses seven zrandom calls on an accepted multi-limb input. The ordinary route zprime(n,8,2) uses base 2 first and seven random bases through ordinary modular exponentiation. Those exact settings were fixed before cohort execution, using the already published seven-output recipe rather than searching labels.

Source call path; dashed connection is proposed



This is a source-supported candidate reason for seven RNG calls before point generation. It does not establish that these were the historical calls. Each zrandom uses rejection sampling, so seven calls need not mean seven state advances; its bounds also differ from a discarded full-limb draw. No new six-base-point reconstruction, caller recovery or independent historical prediction was executed in this study. The connection remains proposed.

For a prime n and a nonzero residue a, write n−1=2^s d with d odd. Fermat's theorem and the two roots of X^2-1 imply the strong probable-prime test accepts. All generated small bases here are below 2^26, far below every original order. Thus rejection would require implementation/input behavior, not the ordinary mathematical test rejecting a true prime. This standard argument is conditional on correct arithmetic; it is not a formal verification of the recovered C. The actual replay checks this recovered configuration on the complete census.

Frozen workload, controls and actual execution

The human protocol and structured version-3 protocol preceded the 84-call cohort. The canonical non-freeze protocol hash is 6cddd67e331ed5ae990c90cfa7c8ee94fdfe00a84a1dccf239fa6fc179dcd0ca. JSON Schema validation and native-source/file bindings are recorded in protocol admission. Labels were already visible; there is no blinded preregistration, p-value or holdout claim. The 600-second field is a conservative enforced resource ceiling, not a measured or forecast algorithm runtime. The original Python protocol checker was not executed; schema/file admission and manual semantic review are named precisely rather than presented as its certification.

Version	Routine	Published accepted	Skipped accepted	Unknown
1.0	zprobprime(n,7)	10/10	11/11	0
1.0	zprime(n,8,2)	10/10	11/11	0
1.1	zprobprime(n,7)	10/10	11/11	0
1.1	zprime(n,8,2)	10/10	11/11	0

Each call starts a fresh process with the unchanged default RNG initialization 7157891. All 21 exact decimal inputs were byte-copied from previously proved original-order packets, with the complete parameter table retained in the parent study. The three originally supplied skipped P-224 curves and skipped P-256 candidate 876 pass every tested configuration. No control was omitted.

The modern compatibility build uses the original arithmetic bodies, Zig 0.15.2 x86-linux-musl, GNU89/O0/fcommon/fwrapv, SINGLE_MUL, 26-bit limbs and DOUBLES_LOW_HIGH. The strict C99 adapter asserts long=4, double=8, char=8 and the IEEE word layout. A typed dependency shim isolates legacy header-private declarations without patching the original library. Undefined-behavior instrumentation is retained with halt_on_error=1. Both versions passed 13 fixed generic prime/composite fixtures through both routes: 52 development observations before the historical cohort. This is a recovered-source modern build, not an authenticated historical executable or compiler/libc replay.

Source commit 02e5bfc524a0a2564d8e353f32c6c9bd06f731d7 launched the immutable script after a successful S3 gate. Serial QEMU i386 7.2.0 calls used a 256 MiB guest reservation, 768 MiB host virtual-address cap, 60-second per-call and 600-second total watchdog, with no retries and disabled core dumps. All 84 processes exited zero. No clock, CPU/rusage measurement, performance isolation, speed comparison, DLP experiment or security conclusion is admitted.

Output correction preserved separately

The initial observer required a single-line decimal root. Original zwriteln uses zfwrite_c, which emits backslash-newline continuations at its line width. Consequently all 12 larger orders per configuration were initially indeterminate: 4 published and 5 skipped were admitted, with no rejection. Every original routine had actually returned 1.

Saved validation 002 corrected only the observer's source-defined continuation grammar after seeing those outputs. It requires the complete requested decimal root, exact ABI/method, one final result and no trailing data. Three positive grammar controls and seven malformed/root-bound controls pass. All 84 saved input/status/stdout/stderr packets are hash-bound before and after validation. No original routine was rerun, and no original input or output changed. The frozen initial source and 48 initial unknown observations remain in the raw archive. This is a disclosed post hoc I/O correction, not a new hypothesis or 84 new samples.

The unified native export has 84 planned, attempted and completed calls, zero incomplete/interrupted/resumed/excluded. Unmeasured host, clock, rusage and execution-time fields remain null. Its verification concerns the input-bound software observation, not a new mathematical prime certificate. The corrected analysis hash is explicitly different from the initially frozen observer. Export admission checks structure and all referenced bytes; it does not claim the canonical Python checker ran.

Backend and development failures

Current repository policy defaults new CPU experiments to EKS Spot. The actual configured account matches, but CodeBuild receipt reads and EKS cluster access are denied; no custom-command MCP is installed, and the portable bridge admits the ecbench benchmark contract rather than this legacy command. No Spot job or paid resource was launched. A prospective amendment records the bounded local fallback within the existing authorized native-analysis workflow, with the same inputs/controls and no timing claim. No IAM, network policy or service protection was broadened, and no unrelated fixture substituted for the task.

Development outcome	Attempts
Strict driver exposed legacy header warnings/private declarations	5
Unadmitted LP64 platform configuration timed out on generic fixture 2	2
ILP32 link, executable-mode or missing endian setting failed	3
Final explicit ILP32 configurations passed all fixed fixtures	2

All 12 development attempts are preserved, including every error. The LP64 SINGLE_MUL union has incompatible unsigned-long words; the original configurator requires DOUBLES_LOW_HIGH on x86. These are platform-control failures, not rejected historical curves. The first ILP32 link lacked Zig's O0 UBsan runtime; the runtime was linked rather than disabled. The corrected observer, native exporter and operational metric repair have separate strict build checks. Descriptive reports and PDFs cover development, actual cohort execution and saved validation.

What this changes, and what it leaves open

This excludes the tested FreeLIP versions/configurations as a universal reject-all-skips explanation when supplied the correct original orders. It does not exclude a caller feeding different or incorrect point counts, unsupported historical ABI/compiler behavior, an order-computation failure, a different library/configuration, or another acceptance step. The exact original generation/acceptance program is still missing.

The complete-factor comparison remains 84 proved complete records over all 21 curves. The existing two-test generation and three-test structural fits remain empirical, with their original calibration and root-convention limits. Exact embedding and SEA-proxy tests also remain separate negative evidence. No trait difference or

recovered software behavior establishes an attack on selected or skipped curves. Neither weakness hypothesis is resolved.

Requirement	Status
Original ICV1/DiSSECT assessment and four factorizations	Preserved complete; all 21 controls retained.
Actual FreeLIP full-census replay	Complete: 84/84 calls; every configuration accepts all 21.
Seven-call/base-point relationship	Source-supported proposed mechanism; original caller and exact new reconstruction unverified.
Historical skip criterion	Unresolved; no supported rule explains every skip.
New attack or measured DLP	Not established; no DLP experiment performed.

Checked the canonical IC scoreboard, progress timeline, curve diagrams and performance-gain pages: this functional order-only result supplies no attack, speed or crossover value to promote. Those graphs remain unchanged. Study figures and the neutral distinct-order coverage metric are updated. The existing problem record preserves checkable historical-evidence and attack obligations.

Evidence custody and reproduction

The preparation archive was stored and fully downloaded before any historical call: 417 source/development/prepared-package files verified. The final archive contains the original 84 execution packets, corrected saved validation, initial/final summaries, snapshots and native exporter. Preparation receipt and final receipt bind complete bytes and every manifest member. No credentials or whole workspace were archived. The reproduction guide explains hydration, extraction and hashes. Current PDFs/PNGs are stored with full GET verification; old studies stay frozen.